llmXive Automated Review of Leveraging Verifier-Based Reinforcement Learning in Image Editing

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the alignment between claims and their cited sources.

1. Dataset Attribution and Methodology (Section 3.1): The manuscript states, "We curate 200K samples from Imgedit... 100K Hard (curated via GPT-4o)." The citation \citep{ye2025imgedit} refers to the dataset paper. It is unclear if the "Hard" subset is a standard part of the *Imgedit* benchmark or a novel filtering step performed by the authors. If the latter, the phrasing implies the dataset inherently contains this GPT-4o filtered split, which may be inaccurate. The claim should clarify that the authors applied a GPT-4o filter to the base Imgedit dataset.

2. Numerical Precision and Baseline Consistency (Abstract, Section 4.2, Table 1): The text reports the 7B model's accuracy as **82.22%**, while Table 1 lists it as **82.2%**. This inconsistency in precision should be resolved. Furthermore, the claim that the model "surpasses Seed-1.5-VL" compares 82.2% against Seed-1.5-VL's **79.3%** (the "T+V" column). However, Table 1 shows Seed-1.5-VL achieving only **72.2%** in the "T" (Think) only column. The text must explicitly state that the comparison is against the *best-performing configuration* (T+V) of the baseline to ensure the claim of superiority is not misleading.

3. Attribution of Decomposition Principles (Section 3.1): The text states: "Seed-1.5-VL decomposes tasks into principles: (a) Keep, (b) Follow, (c) Quality." Unless the Seed-1.5-VL paper explicitly defines these three specific categories as its decomposition strategy, the authors are attributing a specific prompt engineering or internal logic to an external model. If this decomposition was designed by the authors *using* Seed-1.5-VL as a tool, the phrasing should be "We employ Seed-1.5-VL to decompose tasks..." rather than implying Seed-1.5-VL inherently possesses this logic.

4. Benchmark Terminology (Section 4.2): The text claims performance on "EditRewardBench" citing \citep{luo2025editscore}. The authors must ensure that "EditRewardBench" is the exact name used in the *EditScore* paper. If the benchmark is named differently in the source, the claim should match the source terminology to avoid confusion.

Conclusion: The core scientific claims appear supported by the provided tables, but the textual descriptions occasionally blur the line between the authors' contributions and the capabilities of cited external models/datasets. Clarifying these attributions and ensuring consistent numerical reporting will strengthen the accuracy of the claims.

Action items.

- [**writing**] Clarify if the 'Hard' subset of Imgedit (Sec 3.1) is an inherent dataset feature or a new GPT-4o filter by authors to avoid misattributing methodology to the cited source.
- [**writing**] Ensure the '82.22%' accuracy claim matches Table 1's '82.2%' and explicitly state the comparison is against the Seed-1.5-VL 'T+V' baseline, not the 'T' baseline.
- [**writing**] Verify if 'EditRewardBench' is the exact name in the cited EditScore paper (Sec 4.2) or if the terminology needs adjustment to match the source.
- [**writing**] Clarify if the 'Keep, Follow, Quality' principles (Sec 3.1) are inherent to Seed-1.5-VL

or a prompt design by the authors to avoid misattribution.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 effectively illustrates the framework and benchmark results, but the downstream application radar chart (1c) lacks a numerical scale for interpretation, and the caption contains a typo in the model name.

Figure 2

The figure effectively visualizes the input components for the RRM, but the caption's reference to a 'quadruple' is confusing given the visual layout, and the text block containing the principles is too small to be easily legible.

Figure 3

The figure effectively demonstrates the input quadruple concept but suffers from a caption mismatch regarding the inclusion of the edited image and has poor legibility for the principle text.

Figure 4

The figure presents a qualitative comparison of image editing results, but the bottom row contains significant errors where the displayed edits do not match the provided instruction captions (a spoon is added instead of a bird, and the zebra is not removed).

Figure 5

The figure presents a grid of qualitative results but fails to visually distinguish between the two datasets (GEdit-Bench and Emu Edit) described in the caption. Additionally, the claimed comparison with the baseline model is not visible in the image, as only the input and final output are shown.

Figure 6

The figure displays a grid of edited images but fails to include the specific edit instructions (prompts) or clear labels distinguishing input from output, making it impossible to verify the claimed 'complex instructions' or robust capabilities.

Figure 7

The figure provides a clear visual comparison of editing results, but the column headers ('Baseline', 'w. Edit-R1') are less explicit than the caption's description ('Qwen-Edit', 'Qwen-Edit w. Edit-R1'), and the examples provided do not fully illustrate the claimed 'fine-grained attribute changes' beyond motion edits.

Figure 8

The figure fails to demonstrate the claimed success of the RL model; the 'Winner' image incorrectly alters the hat color and fails to render the shirt in the requested bright red, while the 'Loser' image actually performs the color change correctly.

Figure 9

The figure provides a clear visual breakdown of the training pipeline, but the caption contradicts the diagram's internal labels regarding the 'Cold Start' phase, and the mathematical notation for preference pairs is undefined in the text.

Figure 10

The figure effectively displays training dynamics with clear legends and axes, but the caption contains a garbled mathematical formula for the weighted advantage that needs correction for clarity.

Figure 11

The figure is a duplicate of Figure 10, but the caption is a raw, commented-out string that incorrectly describes Figure 12's content, resulting in a total disconnect between the visual evidence and the provided text.

Figure 12

The figure displays training and evaluation reward curves clearly, but the caption is incomplete and contains draft artifacts. Additionally, the x-axis label lacks a specific unit definition.
Action items.

- **[writing]** Figure 1c: The legend entry 'FLUX.Kontext (Base)' contains a typo ('batifol2025flux') in the caption text, and the legend label itself is slightly ambiguous regarding the specific model version.

- [**science**] Figure 1c: The radar chart lacks a numerical scale or concentric grid lines indicating the magnitude of the plotted values, making it impossible to visually verify the 'significant improvement' claimed in the caption.
- [**writing**] Figure 2: The caption describes the content as an 'input quadruple' (implying four components), but the visual only explicitly labels three: 'Source Image', 'Instruction', and 'Edited Image'. The 'decomposed principles' listed below are not visually linked to the image flow or labeled as a distinct input component in the diagram.
- [**writing**] Figure 2: The JSON text block labeled 'Principle' is extremely dense and small, making it difficult to read the specific questions and categories without zooming in significantly.
- [**writing**] Figure 3: The caption states the figure consists of a 'source image, edit instruction, and the decomposed principles,' but the rendered image also includes an 'Edited Image' and an 'Instruction' arrow, creating a mismatch between the visual content and the caption description.
- [**writing**] Figure 3: The JSON list under the 'Principle' heading is rendered in a very small font size, making the text difficult to read and illegible at standard viewing sizes.
- [**writing**] Figure 4: The caption for the bottom-left row reads '[Subject Add] Add a robot bird in the sky..', but the visual result shows a spoon added to the bowl, not a robot bird in the sky. This is a mismatch between the instruction text and the displayed edit.
- [**writing**] Figure 4: The caption for the bottom-right row reads '[Subject Remove] erase the zebra.', but the visual result shows the zebra is still present in the rightmost panel (w. Edit-R1), failing to execute the instruction shown in the text.
- [**science**] Figure 5: The caption claims the figure contains two distinct sections, (a) GEdit-Bench comparisons and (b) Emu Edit Test Set results, but the rendered image displays a single grid of 12 examples without any visual separators or labels to distinguish these two datasets.
- [**writing**] Figure 5: The caption states that section (a) shows a comparison between the baseline and the enhanced model, but the image grid only displays pairs of images (Input vs. Result) without showing the baseline model's output for direct comparison.
- [**science**] Figure 6: The caption claims to show results on the 'Emu Edit Test Set' but lacks the specific edit instructions (prompts) for each example. Without the text prompts, the 'robust capabilities' and 'complex instructions' cannot be verified or understood by the reader.
- [**writing**] Figure 6: The figure consists of a grid of images with no internal labels (e.g., 'Input', 'Output') or row/column headers. While the caption implies a comparison, the visual layout does not explicitly distinguish between the source image and the edited result.
- [**science**] Figure 7: The caption claims a comparison between 'Qwen-Edit' and 'Qwen-Edit w. Edit-R1', but the image labels read 'Input image', 'Baseline', and 'w. Edit-R1'. While

'Baseline' is defined as Qwen-Edit in the caption, the specific label 'w. Edit-R1' is ambiguous without the explicit model name 'Qwen-Edit' in the figure header, potentially confusing readers if this figure is viewed in isolation.

- [**science**] Figure 7: The caption states the figure shows results for 'motion-related edits' and 'fine-grained attribute changes', but the visual examples (dog, person, baby, cat) are generic and do not explicitly demonstrate 'fine-grained attribute changes' (e.g., texture, material) distinct from the motion changes shown, making the claim of 'diverse set' slightly overstated for the specific examples provided.
- [**science**] Figure 8: The 'Winner' image fails to follow the instruction to change the shirt to red; it displays a dark maroon shirt, whereas the 'Loser' image correctly displays a bright red shirt. This contradicts the caption's claim that the 'Winner' output correctly executes the edit.
- [**science**] Figure 8: The 'Winner' image incorrectly changes the hat color from blue to light blue, contradicting the caption's claim that it 'correctly preserves the blue hat'.
- [**writing**] Figure 9: The caption describes the bottom section as 'GCPO' (Group Contrastive Preference Optimization), but the diagram explicitly labels the final stage as 'SFT Data Construction for RRM Cold Start', creating a contradiction between the text description and the visual flow.
- [**writing**] Figure 9: The bottom section labels the input as 'Human Annotation: x^l < x^w' and 'Win/Loss Sample', but the caption text does not explicitly define the symbols 'x^l' (loser) and 'x^w' (winner) or explain the preference pair notation.
- [**writing**] Figure 10: The caption contains a likely typo in the mathematical definition of weighted advantage ($1G_i=1^GA_iL_i$), which appears garbled and does not match standard notation or the visual data.
- [**writing**] Figure 10: The caption defines the weighted advantage formula but does not explicitly define the 'Weighted Advantage' metric shown in panel (c) in the context of the GCPO phase, relying on the user to infer the connection to the formula.
- [**writing**] Figure 11: The caption is formatted as a comment block (starting with '%' symbols) and appears to be a raw copy-paste of Figure 10's caption, failing to describe the actual content of the provided image.
- [**science**] Figure 11: The provided image is identical to Figure 10 (Training dynamics of RRMs), yet the caption claims to describe 'Training dynamics of editing model optimization with different RRMs' (which matches the description for Figure 12), creating a complete mismatch between the visual data and the text.
- [**writing**] Figure 12: The caption text is incomplete, ending abruptly with 'acts as a stricter'.
- [**writing**] Figure 12: The caption contains raw formatting artifacts (e.g., '%') and appears to be a draft version rather than a finalized description.

- **[science]** Figure 12: The x-axis label 'RM' is ambiguous and does not explicitly define the unit (e.g., 'Training Steps' or 'Epochs'), making the time scale difficult to interpret.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on a dense layer of specialized acronyms and coined terms that are not consistently defined for a general computer vision or machine learning audience. While the core concepts (verifier, reward model, reinforcement learning) are standard, the specific naming conventions used here create unnecessary friction.

First, the acronym **RRM** (Reasoning Reward Model) is central to the paper's contribution but is introduced in the Abstract as "Edit-RRM" without explicitly spelling out "Reasoning Reward Model" in that same sentence. The full expansion appears later in the Introduction. This forces the reader to guess the meaning of the acronym before it is defined. Similarly, **GCPO** (Group Contrastive Preference Optimization) is introduced in the Abstract and Introduction without the full phrase preceding the acronym, assuming the reader is already familiar with this specific algorithmic variant.

Second, the **GSB** protocol mentioned in the Appendix (Section "Human Evaluation") is presented with a formula $(G-B)/(G+S+B)$ but lacks a definition of the variables G, S, and B. While likely "Greater," "Same," and "Better," this is not stated, making the metric opaque to non-experts.

Third, the term **quadruple** is used frequently (e.g., Section 3.1) to describe the data structure $(x_{\mathrm{edit}}, x_{\mathrm{ref}}, q, \mathcal{P})$. While mathematically precise, a brief parenthetical explanation (e.g., "a four-tuple of...") would improve accessibility.

Finally, **Flow-GRPO** is cited as a specific method in Section 4.1 without explaining the "Flow" component (presumably Flow Matching) or how it modifies the standard GRPO algorithm. This assumes a level of domain-specific knowledge that excludes readers from adjacent fields.

To improve readability, the authors should ensure every acronym is defined at its first occurrence in the text (Abstract, Introduction, or Method) and consider replacing highly specific coined terms with more descriptive phrases where possible.

Action items.

- **[writing]** The manuscript relies heavily on a dense layer of specialized acronyms and coined terms that are not consistently defined for a general computer vision or machine learning audience. While the core concepts (verifier, reward model, reinforcement learning) are standard, the specific naming conventions used here create unnecessary friction. First, the acronym RRM (Reasoning Reward Model) is central to the paper's contribution but is introduced in the Abstract as "Edit-RRM" without explicitly spelli

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a coherent framework for Edit-R1, but several logical gaps exist in the derivation of the proposed algorithms and the justification of their novelty.

First, the mathematical formulation of the Group Contrastive Preference Optimization (GCPO) in Section 3.1.2 (Method) is logically incomplete. The text introduces the win/loss ratios (r^w_j, r^l_j) in Equation 1 but fails to explicitly define the advantage function A^w_j and A^l_j in the main text. While the Appendix mentions "Advantages are computed within each group," the specific transformation from the raw ratios to the advantage signal used in the clipped surrogate loss (Eq. 2) is omitted. Without this explicit definition, the reader cannot verify if the proposed objective function logically follows from the stated win/loss ratio mechanism.

Second, the claim of novelty regarding GCPO requires stronger logical support. The paper asserts that standard algorithms like DPO are "ill-suited" for this task (Introduction). However, DPO is fundamentally designed to optimize policies using pairwise preference data without requiring a separate reward model training step. The paper does not logically demonstrate why a standard DPO formulation cannot be applied to the reasoning traces of the RRM, nor does it explain why the specific "group contrastive" mechanism is necessary over standard pairwise contrast. The distinction between GCPO and existing pairwise RLHF methods remains a claim rather than a derived necessity.

Finally, the logical flow of the two-stage training pipeline (Cold-Start SFT followed by GCPO) lacks a clear justification for the transition. The SFT stage uses an external VLM (SeedVLM-1.5) to select "highest accuracy" CoT traces. The paper does not logically address how this selection process avoids propagating the external VLM's specific biases into the RRM before the GCPO stage begins. If the SFT data is biased, the GCPO stage must logically overcome this, but the paper does not provide a mechanism or theoretical argument for how GCPO corrects for potential SFT-induced bias, leaving a gap in the causal chain of the training process.

Action items.

- [**writing**] The manuscript presents a coherent framework for Edit-R1, but several logical gaps exist in the derivation of the proposed algorithms and the justification of their novelty. First, the mathematical formulation of the Group Contrastive Preference Optimization (GCPO) in Section 3.1.2 (Method) is logically incomplete. The text introduces the win/loss ratios (r^w_j, r^l_j) in Equation 1 but fails to explicitly define the advantage function A^w_j and A^l_j in the main text. While the Appendix m

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The manuscript makes several claims that extend beyond the immediate evidence presented in the text and tables, specifically regarding novelty and comparative performance.

First, the assertion of novelty in Section 2.1 and Table 1 is overreaching. The authors state that Edit-RRM is "uniquely positioned" and the "first" to integrate principle-decomposition CoT with a two-stage training pipeline for visual editing tasks. However, Table 1 explicitly lists "VisualQuality-R1" and "EditScore" as concurrent works. While the authors mark Edit-RRM with checkmarks for all three reasoning features, the text fails to sufficiently distinguish how these concurrent works differ or why they do not qualify as "firsts" in the specific context claimed. The claim of being the "first" is too absolute given the presence of these cited contemporaries in the same table.

Second, the performance claims in the Abstract and Section 4.2 regarding "Seed-1.6-VL" are unsupported. The text states the 7B model surpasses both "Seed-1.5-VL and Seed-1.6-VL." However, Table 1 ("Accuracy on our internal benchmark") only provides a row for Seed-1.5-VL (79.3%). There is no data row or reference to Seed-1.6-VL's performance in the provided tables. Claiming a specific numerical superiority over a model for which no data is presented in the results section is a clear over-claim.

Finally, the interpretation of the GCPO results in Section 4.2 ("Impact of GCPO and Scalability") over-interprets the data. The authors claim GCPO "transforms the RRM into a stricter evaluator" because evaluation rewards are higher while training rewards are lower. While this observation is interesting, the paper presents this as a definitive causal mechanism without ruling out alternative explanations (such as overfitting to the specific distribution of the evaluation set or a shift in the reward landscape that doesn't necessarily imply "strictness" in a general sense). The language used ("transforms," "yielding higher evaluation rewards") suggests a robust, proven mechanism that the current data (a single observation of diverging trends) does not fully justify.

Action items.

- [**science**] The claim that Edit-RRM is the 'first' generative, pointwise verifier for image editing (Section 2.1, Table 1) is overreaching. Concurrent work like EditScore (Luo et al., 2025) and VisualQuality-R1 (Wu et al., 2025) are cited in the bibliography and table, yet the text asserts uniqueness without addressing their specific architectures or distinguishing features sufficiently to support the 'first' claim.
- [**science**] The abstract and Section 4.2 claim the 7B model 'surpasses Seed-1.5-VL and Seed-1.6-VL' with 82.22% accuracy. However, Table 1 only lists Seed-1.5-VL (79.3%) and does not provide data for Seed-1.6-VL. The claim regarding Seed-1.6-VL is unsupported by the provided data tables and constitutes an over-claim.

- [**writing**] The paper claims GCPO 'transforms the RRM into a stricter evaluator' (Section 4.2) based on the observation that evaluation rewards are higher while training rewards are lower. This causal link is asserted without statistical justification or a discussion of potential confounding factors (e.g., distribution shift), over-interpreting the correlation as a definitive mechanism.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a novel framework, Edit-R1, for image editing using a verifier-based reasoning reward model. From a safety and ethics perspective, the paper is generally sound but requires specific clarifications regarding human subject research and data provenance before acceptance.

Human Subjects and IRB Compliance The methodology section (Section 3.1.2, "Preference Data") states that the authors constructed a dataset of approximately 10,000 human-annotated preference pairs. However, the manuscript does not include an Institutional Review Board (IRB) approval statement, nor does it detail the informed consent process, participant compensation, or ethical oversight mechanisms. Given that human annotators were involved in generating the core training signal for the reward model, this is a critical omission. The authors must add a statement confirming that the study was conducted in accordance with relevant ethical guidelines and that informed consent was obtained from all participants.

Data Provenance and Licensing The training pipeline involves curating 200K samples from the "Imgedit" benchmark (Section 3.1.1) and using external VLMs (Seed-1.5-VL) to generate and select reasoning traces. The authors must explicitly clarify the licensing terms of the Imgedit dataset and confirm that the use of these samples for training a new model is permitted under the original license. Furthermore, the reliance on proprietary or external VLMs (Seed-1.5-VL) to generate the "cold-start" SFT data raises questions about the reproducibility and potential copyright implications of using VLM-generated reasoning traces as ground truth. A brief discussion on the legal and ethical status of these derived data points is recommended.

Dual-Use and Safety Mitigation The paper focuses on improving instruction following and reducing hallucinations in image editing. While this is a positive step, the technology inherently carries dual-use risks. A more robust reward model could potentially be used to refine models for generating highly realistic deepfakes or to bypass existing safety filters in generative models. The authors should include a brief discussion in the Conclusion or a dedicated "Limitations and Safety" section acknowledging these risks and describing any specific measures taken (e.g., filtering training data for sensitive content, implementing safety classifiers) to prevent misuse.

Conclusion The scientific contribution is significant, but the manuscript must address the ethical oversight of human data collection and clarify data licensing to meet publication standards.

Action items.

- [**writing**] The manuscript describes a human annotation pipeline for ~10,000 preference pairs (Sec 3.1.2) but lacks an IRB statement or explicit description of informed consent procedures, compensation, and ethical oversight. This is required for publication involving human subjects.
- [**writing**] The data generation pipeline relies on external VLMs (Seed-1.5-VL) to curate 200K samples and select CoT trajectories (Sec 3.1.1). The authors must clarify the data provenance, licensing status of the 200K samples from Imgedit, and whether the VLM outputs used for training constitute copyrighted material or require specific attribution beyond citation.
- [**writing**] The paper proposes a verifier-based reward model for image editing. While the intent is to improve quality, the authors should briefly address potential dual-use risks, such as the model being used to generate deceptive deepfakes or bypass safety filters in downstream editing models, and describe any mitigation strategies employed.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a novel framework, Edit-R1, for image editing using a verifier-based reasoning reward model (RRM). While the methodological approach of decomposing instructions into principles and using a two-stage training pipeline (SFT + GCPO) is sound, the scientific evidence supporting the central claims lacks necessary statistical rigor and transparency regarding experimental setup.

First, the primary quantitative claim in Table 1 (Section 4.2) states that the 7B Edit-RRM achieves 82.2% accuracy, surpassing Seed-1.5-VL (79.3%). However, the text and table provide no measure of variance (e.g., standard deviation, confidence intervals) across multiple runs or bootstrap samples. Given the relatively small margin of improvement (~2.9%), it is impossible to determine if this gain is statistically significant or within the noise floor of the evaluation metric. Furthermore, the composition of the "internal benchmark" (5,000 samples) is not described in sufficient detail to assess potential selection bias or domain shift.

Second, the human evaluation results presented in the Appendix (Section "Human Evaluation") report a single GSB score of +23.2. This is a critical result for validating the downstream impact of the RRM, yet the manuscript fails to report the sample size (number of image pairs evaluated), the number of human annotators, or any measure of inter-annotator agreement. Without these details, the reliability of the +23.2 figure cannot be verified, and the claim of "substantial enhancement" remains anecdotal rather than empirical.

Finally, the training data for the GCPO stage consists of approximately 10,000 human preference pairs. The paper does not discuss the statistical power of this dataset size or provide ablation studies showing how performance scales with the number of preference pairs. Given the complexity of the reasoning task, there is a risk that the model is overfitting to a small, potentially non-representative subset of human preferences. The absence of error bars, confidence intervals, or robustness checks across different data splits weakens the evidentiary support for the proposed method's superiority.

Action items.

- [**science**] The paper claims 82.22% accuracy for the 7B RRM on an internal benchmark (Tab. 1) but does not report confidence intervals, standard deviations, or the specific composition of the 5,000-sample test set. Without variance metrics or a clear description of the test distribution, the statistical significance of the ~3% gain over Seed-1.5-VL (79.3%) cannot be assessed.
- [**science**] The human evaluation section (Appendix) reports a single GSB score of +23.2 for FLUX.Kontext but omits the sample size (N), the number of human annotators, and the inter-annotator agreement (e.g., Krippendorff's alpha). A single point estimate without error bars or sample details is insufficient to validate the claimed downstream improvement.
- [**science**] The GCPO training utilizes ~10,000 human preference pairs, which is a small fraction of the 200K SFT data. The paper does not provide an ablation study or statistical analysis demonstrating that this specific subset size is sufficient to prevent overfitting or that the performance gains are robust to variations in the preference data selection.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a novel framework for image editing, but the statistical rigor of the reported results requires clarification to ensure the claims are robust.

First, regarding the Reward Model performance in **Table 1** (Section 4.2), the authors report point estimates for accuracy (e.g., 82.2% for the 7B model vs. 79.3% for Seed-1.5-VL). Without confidence intervals (CIs) or standard errors derived from the 5,000-sample test set, it is impossible to determine if the observed ~3% improvement is statistically significant or within the margin of error. Given the large sample size, a binomial proportion confidence interval should be calculated and reported to substantiate the claim of "surpassing" baselines.

Second, the **Human Evaluation** results in the Appendix (Section "Human Evaluation") present a single GSB score of +23.2. The text does not specify the number of human annotators, the number of image pairs evaluated, or the statistical test used to derive this aggregate score. A single point estimate without a measure of variance (e.g., standard deviation across annotators or a p-value from a sign test) limits the reproducibility and reliability of this claim.

Third, the **Reinforcement Learning** methodology in **Equation 2** (Section 3.3) defines the advantage A_i using a normalization term involving the standard deviation of the rewards within a group of G images. The text does not specify whether this standard deviation is computed per-batch or across the entire training run, nor does it detail the value of ϵ_{std}. In RLHF, the choice of normalization and the handling of zero-variance cases are critical for training stability. The authors should explicitly state the normalization scope and the value of the epsilon term to ensure the experiment can be reproduced.

Finally, while the paper mentions scaling from 3B to 7B parameters, it lacks a formal analysis of the scaling law or the statistical significance of the performance gap between these model sizes. A brief discussion on the variance of performance across different random seeds (if applicable) would strengthen the evidence for the claimed scalability.

Action items.

- [**science**] Report confidence intervals or standard errors for the accuracy metrics in Table 1 (e.g., 82.2% vs 79.3%) to establish statistical significance of the improvement over baselines.
- [**science**] Clarify the statistical methodology for the GSB score (+23.2) in the Human Evaluation section. Specify the sample size (N) and the statistical test used to derive this metric.
- [**science**] Define the normalization procedure for the advantage calculation in Eq. 2. Explicitly state if the standard deviation is computed per-batch or globally, and how epsilon is chosen to prevent instability.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript presents a compelling framework, but the writing quality requires specific attention to structural flow and minor mechanical errors before publication.

The most significant issue is the structural duplication of the "Related Works" section. The text provided in chunk `e001` contains a "Related Works" section (Section 2) that is immediately followed by "Method" and "Experiments." However, chunk `e002` (which appears to be the Introduction or a continuation) also contains a "Related Works" section with different content and ordering. This suggests a copy-paste error or a failure to merge drafts, which disrupts the logical flow of the paper. These sections must be consolidated into a single, coherent narrative.

On a sentence level, there is a clear typo in the header of the Related Works section in `e001`: "Related Works" is misspelled as "Related Works" (specifically, the subsection header reads `\subsection{Reward model for generative models}` but the main section header in the text body is labeled `\section{Related Works}` with the label `sec:realted`, and the text itself contains the typo "realted" in the label or header context). Specifically, the label

`\label{sec:realted}` contains a typo ("realted" instead of "related"), which should be corrected for consistency.

Additionally, the bibliography file (`main.bib`) contains a truncated entry for `guo2024real`, ending abruptly at `author=`. While this is a metadata issue, it reflects on the overall polish of the submission and must be fixed to ensure the paper compiles without warnings or missing references.

Finally, terminology consistency needs improvement. The term "Chain-of-Thought" is capitalized in the Introduction but referred to as "think" or "COT" in figure captions and method descriptions (e.g., "thinking COT" in Figure 2 caption). Standardizing this to "Chain-of-Thought (CoT)" upon first use and "CoT" thereafter would enhance readability and professionalism.

Addressing these structural and mechanical issues will significantly improve the manuscript's clarity and readiness for review.

Action items.

- [**writing**] Correct the typo in the Related Works section header: 'realted' should be 'related' (Section 2, e001).
- [**writing**] Resolve the structural inconsistency where the 'Related Works' section appears twice (once in e001 and again in e002) with different content ordering. Merge these into a single, cohesive section to improve flow.
- [**writing**] Fix the incomplete sentence in the bibliography file (main.bib) where the entry for 'guo2024real' is truncated at 'author='.
- [**writing**] Standardize the capitalization of 'Chain-of-Thought' (CoT). It appears as 'Chain-of-Thought' in the Introduction but 'think' or 'COT' in other sections (e.g., Figure captions). Ensure consistent terminology throughout.